KAMESH B

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PROFESSIONAL PROFILE

Mechanical Engineering undergraduate with hands-on experience in CAE and FEA. Skilled in ANSYS and HyperMesh, with practical experience in preparing and analyzing engineering models for simulation. Applied CAE techniques to impact and collision simulations and developed basic proficiency in CAD tools. Strong foundation in mechanical engineering with an interest in CAE/CAD and simulation-driven engineering. Seeking an entry-level CAE/CAD Engineer role to develop expertise in engineering simulation.

TECHNICAL SKILLS

- **CAE & FEA Tools:** ANSYS, HyperMesh
- **CAD & Design Tools:** AutoCAD (Basic), NX CAD (Basic), SOLIDWORKS (Basic)
- **CAE & FEA:** Geometry Preparation & Cleanup, 2D/3D Meshing, Boundary Conditions, Stress/Strain/Deformation Analysis, Post-Processing, Material Property Assignment.
- **Mechanical Engineering:** Engineering Mechanics, Machine Design, Strength of Materials, Manufacturing Processes.
- **Programming Languages:** C, Java

ENGINEERING PRACTICES

- Systematic approach to simulation setup and result evaluation.
- Attention to model accuracy, quality, and engineering requirements.
- Analytical problem solving and engineering decision making
- Application of engineering principles to practical simulation problems.

EDUCATION

2023 - 2027
Bachelor of Engineering - Mechanical Engineering
Knowledge Institute of Technology
CGPA: 7.5

2022 - 2023
Sri Sarada Balamandir Boys Matric Higher Secondary School
(48.5% Secured)

TECHNICAL PROJECTS

Impact & Collision Analysis - HyperMesh :

- Performed **car-wall crash analysis** to study structural response under impact loading.
- Conducted **projectile-wall impact analysis** to evaluate stress, strain, and deformation during high-velocity impact.
- Simulated **ball-bottle impact** to analyze deformation and stress response during collision.
- Performed **table-tennis ball and bat impact analysis** to study the dynamic response during contact.

Maritime Rescue Bot :

- Designed and developed a remotely operated water rescue bot with a lightweight C-shaped floating structure for stability and victim support.
- Integrated DC motors, propellers, battery, and wireless remote-control system for propulsion and maneuvering.
- Built and tested the prototype, identifying challenges in stability,motor alignment, waterproofing, and maneuverability.

INTERNSHIP EXPERIENCE

LAKSHMI PIPE - Industrial Training
Gained practical exposure to raw material handling, identification, storage and preparation processes involved in PVC pipe manufacturing.
Machine Learning Intern – Dev Technology Solution
Explored machine learning concepts through basic tasks and examples.
Worked with datasets to analyze patterns and understand model outcomes.

ACHIEVEMENTS & ACTIVITIES

- Presented a technical paper on “Rescue Bot” at Knowledge Institute of Technology.
- Participated in a hands-on Drone Technology workshop at Kongu Engineering College, gaining practical exposure to drone components, operation, and basic flight principles.